

Cost Accounting

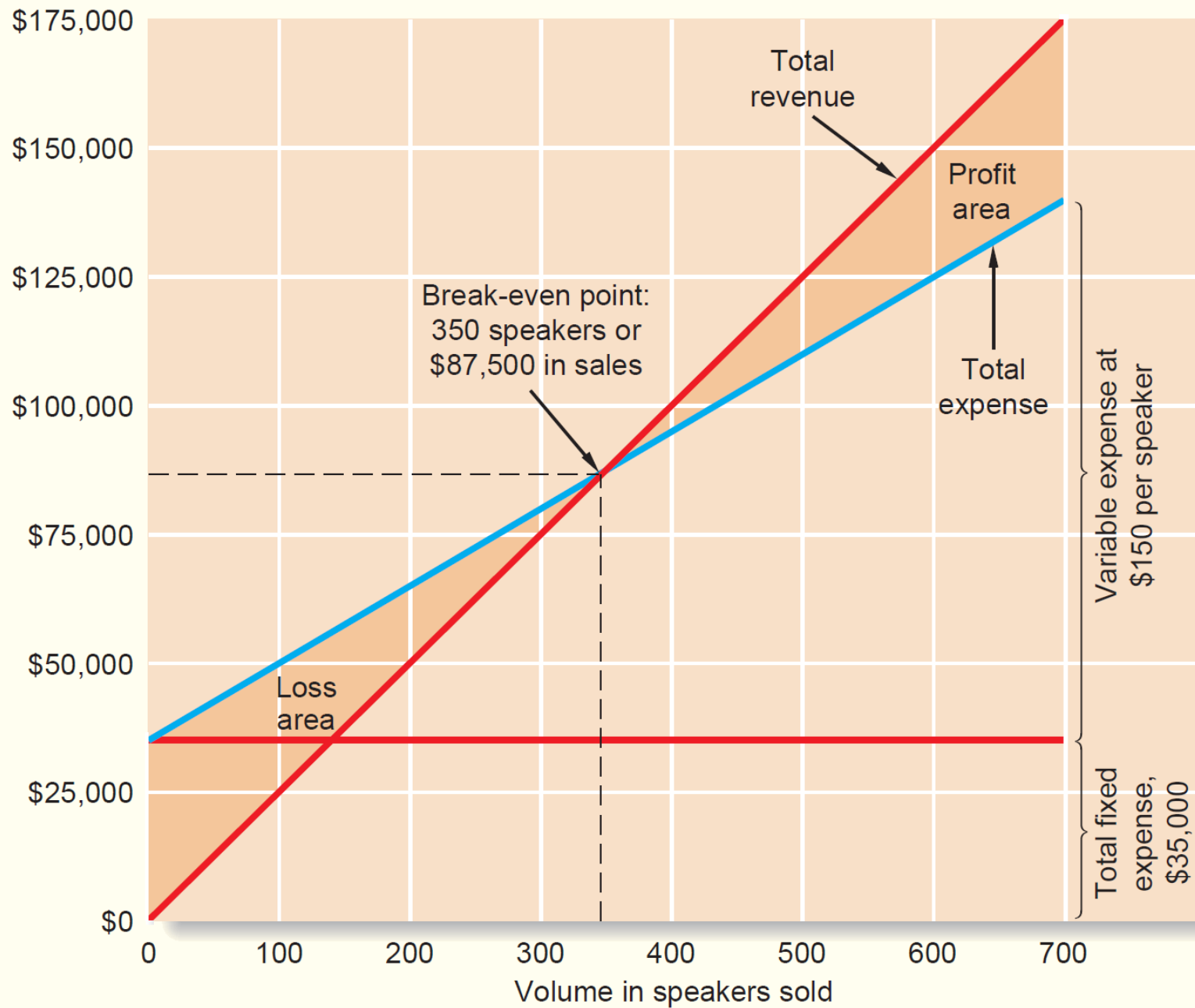
- **Managerial Accounting**
 - *Thirteenth Edition*
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- ***Cost-Volume-Profit (CVP) Relationships***
 - ***Ch.26***

- **Cost-volume-profit (CVP) analysis** is a powerful tool that helps managers understand the relationships among cost, volume, and profit. CVP analysis
 - focuses on how profits are affected by the following factors:
 - 1. Selling prices.
 - 2. Sales volume.
 - 3. Unit variable costs.
 - 4. Total fixed costs.

EXHIBIT 6-2

The Completed CVP Graph



Acoustic Concepts, Inc.
Contribution Income Statement
For the Month of June

	Total	Per Unit
Sales (400 speakers)	\$100,000	\$250
Variable expenses	60,000	150
Contribution margin	40,000	\$100
Fixed expenses	35,000	
Net operating income	\$ 5,000	

Contribution Margin

Contribution margin is the amount remaining from sales revenue after variable expenses have been deducted. Thus, it is the amount available to cover fixed expenses and then to provide profits for the period. Notice the sequence here—contribution margin is used *first* to cover the fixed expenses, and then whatever remains goes toward profits. If the contribution margin is not sufficient to cover the fixed expenses, then a loss occurs for the period.

If company produce 1 speaker what its profit / loss?

- **Contribution margin ratio (CM ratio)** This ratio is computed as follows:
 - **CM ratio = Contribution margin / Sales**
 - CM ratio = Total contribution margin / Total sales
$$\$40,000 / \$100,000 \times 100 = 40\%$$
 - CM ratio = Unit contribution margin / Unit selling price
$$\$100 / \$250 \times 100 = 40\%$$
- The CM ratio shows how the contribution margin will be affected by a change in total sales. Acoustic Concepts' CM ratio of 40% means that for each dollar increase in sales, total contribution margin will increase by 40 cents (\$1 sales CM ratio of 40%). Net operating income will also increase by 40 cents, assuming that fixed costs are not affected by the increase in sales.

- As this illustration suggests, *the impact on net operating income of any given dollar change in total sales can be computed by simply applying the CM ratio to the dollar change*. For example, if Acoustic Concepts plans a \$30,000 increase in sales during the coming month, the contribution margin should increase by \$12,000 (\$30,000 increase in sales $\times$ CM ratio of 40%). As we noted above, net operating income will also increase by \$12,000 if fixed costs do not change. This is verified by the following table:

	Sales Volume			Percent of Sales
	Present	Expected	Increase	
Sales	\$100,000	\$130,000	\$30,000	100%
Variable expenses	60,000	78,000*	18,000	60%
Contribution margin	40,000	52,000	12,000	40%
Fixed expenses	35,000	35,000	0	
Net operating income	\$ 5,000	\$ 17,000	\$12,000	

*\$130,000 expected sales $\div$ \$250 per unit = 520 units. 520 units $\times$ \$150 per unit = \$78,000.

- The relation between profit and the CM ratio can also be expressed using the following
- equation:
 - Profit = CM ratio x Sales - Fixed expenses
 - For example, at sales of \$130,000, the profit is expected to be \$17,000 as shown below:
 - Profit = CM ratio x Sales - Fixed expenses
 - $0.40 \times \$130,000 - \$35,000$
 - $\$52,000 - \$35,000 = \$17,000$

- **Change in Fixed Cost and Sales Volume** Acoustic Concepts is currently selling 400 speakers per month at \$250 per speaker for total monthly sales of \$100,000. The sales manager feels that a \$10,000 increase in the monthly advertising budget would increase monthly sales by \$30,000 to a total of 520 units. Should the advertising budget be increased? The following table shows the financial impact of the proposed change in the monthly advertising budget:

	Current Sales	Sales with Additional Advertising Budget	Difference	Percent of Sales
Sales	\$100,000	\$130,000	\$30,000	100%
Variable expenses	60,000	78,000*	18,000	60%
Contribution margin	40,000	52,000	12,000	40%
Fixed expenses	35,000	45,000†	10,000	
Net operating income	\$ 5,000	\$ 7,000	\$ 2,000	

*520 units × \$150 per unit = \$78,000.
†\$35,000 + additional \$10,000 monthly advertising budget = \$45,000.

• Alternative Solution 1

Expected total contribution margin:	
\$130,000 × 40% CM ratio	\$52,000
Present total contribution margin:	
\$100,000 × 40% CM ratio	<u>40,000</u>
Incremental contribution margin	12,000
Change in fixed expenses:	
Less incremental advertising expense	<u>10,000</u>
Increased net operating income	<u><u>\$ 2,000</u></u>

Alternative Solution 2

Incremental contribution margin:	
\$30,000 × 40% CM ratio	\$12,000
Less incremental advertising expense	<u>10,000</u>
Increased net operating income	<u><u>\$ 2,000</u></u>

IN BUSINESS

GROWING SALES AT AMAZON.COM

Amazon.com was deciding between two tactics for growing sales and profits. The first approach was to invest in television advertising. The second approach was to offer free shipping on larger orders. To evaluate the first option, Amazon.com invested in television ads in two markets—Minneapolis, Minnesota, and Portland, Oregon. The company quantified the profit impact of this choice by subtracting the increase in fixed advertising costs from the increase in contribution margin. The profit impact of television advertising paled in comparison to the free “super saver shipping” program, which the company introduced on orders over \$99. In fact, the free shipping option proved to be so popular and profitable that within two years Amazon.com dropped its qualifying threshold to \$49 and then again to a mere \$25. At each stage of this progression, Amazon.com used cost-volume-profit analysis to determine whether the extra volume from liberalizing the free shipping offer more than offset the associated increase in shipping costs.

Source: Rob Walker, “Because ‘Optimism is Essential,’” *Inc.* magazine, April 2004 pp. 149–150.

- The data and recall again that Acoustic Concepts is currently selling 400 speakers per month. To increase sales, the sales manager would like to cut the selling price by \$20 per speaker and increase the advertising budget by \$15,000 per month. The sales manager believes that if these two steps are taken, unit sales will increase by 50% to 600 speakers per month. Should the changes be made? A decrease in the selling price of \$20 per speaker would decrease the unit contribution margin by \$20 down to \$80.

Expected total contribution margin with lower selling price:	
600 speakers × \$80 per speaker	\$48,000
Present total contribution margin:	
400 speakers × \$100 per speaker	40,000
Incremental contribution margin	8,000
Change in fixed expenses:	
Less incremental advertising expense	15,000
Reduction in net operating income	<u>\$ (7,000)</u>

	Present 400 Speakers per Month		Expected 600 Speakers per Month		Difference
	Total	Per Unit	Total	Per Unit	
Sales	\$100,000	\$250	\$138,000	\$230	\$38,000
Variable expenses	60,000	150	90,000	150	30,000
Contribution margin	40,000	<u>\$100</u>	48,000	<u>\$ 80</u>	8,000
Fixed expenses	35,000		50,000*		15,000
Net operating income (loss)	<u>\$ 5,000</u>		<u>\$ (2,000)</u>		<u>\$ (7,000)</u>

*35,000 + Additional monthly advertising budget of \$15,000 = \$50,000.

- **Change in Variable Cost, Fixed Cost, and Sales Volume** Refer to Acoustic Concepts' original data. As before, the company is currently selling 400 speakers per month. The sales manager would like to pay salespersons a sales commission of \$15 per speaker sold, rather than the fl at salaries that now total \$6,000 per month. The sales manager is confident that the change would increase monthly sales by 15% to 460 speakers per month. Should the change be made?

Expected total contribution margin with sales staff on commissions:	
460 speakers × \$85 per speaker	\$39,100
Present total contribution margin:	
400 speakers × \$100 per speaker	40,000
Decrease in total contribution margin	(900)
Change in fixed expenses:	
Add salaries avoided if a commission is paid	6,000
Increase in net operating income	<u>\$ 5,100</u>

	Present 400 Speakers per Month		Expected 460 Speakers per Month		Difference
	Total	Per Unit	Total	Per Unit	
Sales	\$100,000	\$250	\$115,000	\$250	\$15,000
Variable expenses	60,000	150	75,900	165	15,900
Contribution margin	40,000	\$100	39,100	\$ 85	900
Fixed expenses	35,000		29,000		(6,000)*
Net operating income ...	<u>\$ 5,000</u>		<u>\$ 10,100</u>		<u>\$ 5,100</u>

*Note: A *reduction* in fixed expenses has the effect of *increasing* net operating income.

- **Target Profit Analysis in Terms of Sales Dollars**

- Unit sales to attain the target profit = $\text{Target profit} + \text{Fixed expenses} / \text{Unit CM}$
 $= \$40,000 + \$35,000 / \$100$
 750

- The **Break-even** point as the level of sales at which the company's profit is zero. What we call *break-even analysis* is really just a special case of target profit analysis in which the target profit is zero. We can use either the equation method or the formula method to solve for the break-even point, but for brevity we will illustrate just the formula method. The equation method works exactly like it did in target profit analysis. The only difference is that the target profit is zero in break-even analysis.

- Unit sales to break even = $\text{Fixed expenses} / \text{Unit CM}$
 $= \$35,000 / \100
 $= 350$

- **Break-Even in Sales Dollars**

- Dollar sales to attain a target profit = $\text{Target profit} + \text{Fixed expenses} / \text{CM ratio}$
- Dollar sales to break even = $\$0 + \text{Fixed expenses} / \text{CM ratio}$
- Dollar sales to break even = $\text{Fixed expenses} / \text{CM ratio}$
- The break-even point at Acoustic Concepts would be computed as follows:
- Dollar sales to break even = $\text{Fixed expenses} / \text{CM ratio}$

$$\$35,000 / 0.40$$

$$\$87,500$$

- ***The Margin of Safety***

- The **margin of safety** is the excess of budgeted (or actual) sales dollars over the breakeven volume of sales dollars. It is the amount by which sales can drop before losses are incurred. The higher the margin of safety, the lower the risk of not breaking even and incurring a loss. The formula for the margin of safety is:

- Margin of safety in dollars = Total budgeted (or actual) sales - Break-even sales

- The margin of safety can also be expressed in percentage form by dividing the margin of safety in dollars by total dollar sales:

- Margin of safety percentage = Margin of safety in dollars / Total budgeted (or actual) sales in dollars

- The calculation of the margin of safety for Acoustic Concepts is:

Sales (at the current volume of 400 speakers) (a)	\$100,000
Break-even sales (at 350 speakers).	87,500
Margin of safety in dollars (b)	<u>\$ 12,500</u>
Margin of safety percentage, (b) ÷ (a)	<u>12.5%</u>

Thank you